

Subject : MOBILE AND PERVASIVE COMPUTING (01CT0716)**Date : 16-Sep-2025****Time : 2 Hours****Total Marks : 60****Instructions :**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Do not write/sign/indication/tick mark anything other than Enroll No. at a specific place on the question paper.

		Marks	BL	CO
Que.1				
(A)	A Body Area Network uses a set of 5 sensors, each transmitting at 50 kbps data rate to a smartphone hub. The smartphone then forwards the data via a 4G cellular link of 1 Mbps. (a) Calculate the total BAN data rate required. (b) Determine whether the cellular link can handle the aggregated data.	[4]	L5	CO3
(B)	Describe the security and privacy issues in BAN and propose possible solutions.	[6]	L3	CO4
Que.2				
(A)	Analyze the role of pervasive computing in the manufacturing industry. How does it improve efficiency, safety, and decision-making compared to traditional systems?	[8]	L4	CO1
(B)	Explain SyncML and apply its role in achieving data synchronization across devices. Why is data synchronization important in mobile environments?	[4]	L3	CO3
OR				
(A)	Analyze in detail the integration of biometric systems into human-machine interfaces in pervasive environments. Analyse the trade-offs between security, convenience, and privacy.	[8]	L4	CO1
(B)	A mobile user is traveling at 72 km/h in a cellular network where each cell has a radius of 1.5 km. If the average call duration is 4 minutes, calculate how many handoffs the user is expected to experience during the call.	[4]	L5	CO3
Que.3				
(A)	Apply LEACH protocol in a smart agriculture WSN deployment. Analyze how its clustering mechanism improves energy efficiency and data aggregation.	[8]	L4	CO3
(B)	Apply context-aware mobile services to explain how they improve user experience with examples	[6]	L4	CO3
OR				
(A)	Define macro mobility and micro mobility in wireless networks. Explain their differences with suitable real-life examples and scenarios in mobile IP-based systems.	[8]	L4	CO3
(B)	Compare Wi-Fi and WiMAX	[6]	L4	CO3

Que.4				
(A)	Evaluate Wi-Fi and Z-Wave based on their range, power consumption, and typical use cases. Which is more suitable for smart home IoT devices?	[6]	L5	CO3
(B)	Explain the major functions of a smartwatch and google glass.	[4]	L3	CO4
OR				
(A)	Evaluate the suitability of Wi-Fi, and ZigBee for different IoT applications based on power, data rate, and range.	[6]	L4	CO3
(B)	What are the main types of sensors used in BAN?	[4]	L3	CO4
Que.5				
(A)	Analyze the role of Adaptive Modulation and Channel Coding (AMC) in modern wireless communication systems. How does AMC improve spectral efficiency and link reliability under varying channel conditions? Support your answer with real-world examples.	[8]	L4	CO2
(B)	An LTE system has 15 MHz bandwidth. Each RB = 180 kHz, carrying 200 kbps. A cell has 40 users, each needing 300 kbps. (a) Can the cell support all 40 users? (b) If yes, how many RBs remain free? (c) Give one practical implication.	[6]	L5	CO2
OR				
(A)	Design and draw the Cellular network for $i=2$, $j=2$, and $M=3$. Also highlight the cells that use same frequency band.	[8]	L3	CO2
(B)	An LTE system has 20 MHz bandwidth. Each RB = 180 kHz, carrying 200 kbps. A cell has 50 users, each needing 400 kbps. (a) Can the cell support all 50 users? (b) If yes, how many RBs remain free? (c) Justify with one practical implication.	[6]	L3	CO2

---Best of Luck---

MARWADI UNIVERSITY
MU-FOT
INFORMATION AND COMMUNICATION TECHNOLOGY
Semester SEMESTER 7 - Winter

Subject : MOBILE AND PERVASIVE COMPUTING (01CT0716)

Date : 16-Sep-2025

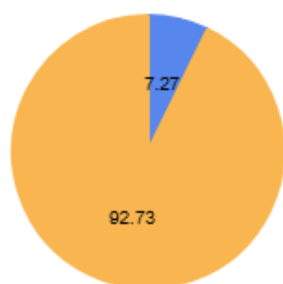
Time : 2 Hours

Total Marks : 60

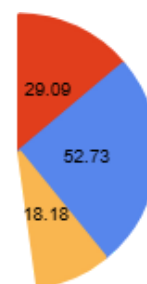
Difficulty Level	Weightage	No of Question	Total Marks	Question List
High	0.00	0	0	
Low	7.27	1	8	5(A)
Medium	92.73	17	102	1(A), 1(B), 2(A), 2(B), 3(A), 3(B), 4(A), 4(B), 5(A), 5(B)

Module Name	Weightage	No of Question	Total Marks	Question List
Overview of Wireless Systems	16.36	3	18	3(B), 4(A)
Medium Access:	14.55	2	16	3(A), 5(A)
Mobility and Handoff Management:	10.91	2	12	2(B), 3(A)
Cellular Networks:	18.18	3	20	5(A), 5(B)
Pervasive Computing:	14.55	2	16	2(A)
Context Aware Sensor Networks:	9.09	2	10	2(B), 3(B)
Recent Advances in Wearable devices	3.64	1	4	4(B)
Body Area Networks (BAN)	12.73	3	14	1(A), 1(B), 4(B)

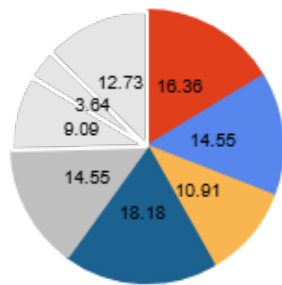
Blooms Taxonomy	Weightage	No of Question	Total Marks	Question List
L1 - Remember / Knowledge	0.00	0	0	
L2 - Understand	0.00	0	0	
L3 - Apply	29.09	6	32	1(B), 2(B), 4(B), 5(A), 5(B)
L4 - Analyze	52.73	8	58	2(A), 3(A), 3(B), 4(A), 5(A)
L5 - Evaluate	18.18	4	20	1(A), 2(B), 4(A), 5(B)
L6 - Higher order Thinking / Creative	0.00	0	0	



● High
 ● Low
 ● Medium



● Apply
 ● Analyze
 ● Evaluate



● Overview of Wireless Systems

● Medium Access:

● Mobility and Handoff Management:

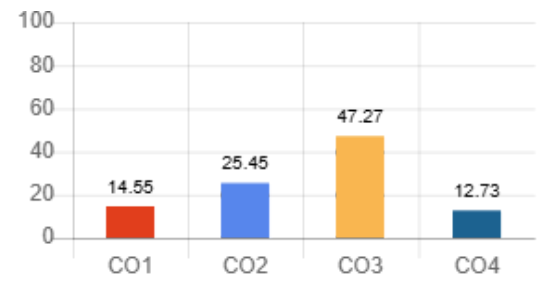
● Cellular Networks:

● Pervasive Computing:

Context Aware Sensor Networks:

Recent Advances in Wearable devices

Body Area Networks (BAN)



Course Outcome Wise Questions

Subject Code	01CT0716	Subject	MOBILE AND PERVASIVE COMPUTING
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CO No.	Course Outcome
CO1	Understand the fundamental concepts of pervasive computing.
	2(A), 2(A-Or)
CO2	Analyze the cellular network architecture and protocols.
	5(A), 5(A-Or), 5(B), 5(B-Or)
CO3	Analyze various wireless network systems.
	1(A), 2(B), 2(B-Or), 3(A), 3(A-Or), 3(B), 3(B-Or), 4(A), 4(A-Or)
CO4	Design and implement pervasive application systems.
	1(B), 4(B), 4(B-Or)

Blooms Taxonomy	Question List
Remember / Knowledge	
Understand	
Apply	1(B), 2(B), 4(B), 4(B-Or), 5(A-Or), 5(B-Or)
Analyze	2(A), 2(A-Or), 3(A), 3(A-Or), 3(B), 3(B-Or), 4(A-Or), 5(A)
Evaluate	1(A), 2(B-Or), 4(A), 5(B)
Higher order Thinking / Creative	